# Quantum Field Theory
# Part II: Dirac and the Spinor

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**Abstract**

This document contains derivations of equations from [1, Part II: Dirac and the Spinor].

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## 2.1 The Dirac Equation

$$
\begin{aligned}
(i\gamma^\mu\partial_\mu - m)\psi =& 0\\
&\implies\\
(i\gamma^\mu\partial_\mu + m)(i\gamma^\nu\partial_\nu - m)\psi =& 0\\
&\iff\\
\big(-\gamma^\mu\partial_\mu\gamma^\nu\partial_\nu + \cancel{mi\gamma^\nu\partial_\nu} - \cancel{mi\gamma^\mu\partial_\mu} - m^2\big)\psi =& 0\\
&\iff\\
\big(\gamma^\mu\partial_\mu\gamma^\nu\partial_\nu + m^2\big)\psi =& 0\\
&\iff\\
\big(\gamma^\mu\gamma^\nu\partial_\mu\partial_\nu + m^2\big)\psi =& 0\\
&\iff\\
\big[(\frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)\partial_\mu\partial_\nu + m^2\big]\psi =& 0\\
\text{Define}\{\gamma^\mu,\gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu.& \iff\\
\big(\frac{1}{2}\{\gamma^\mu,\gamma^\nu\}\partial_\mu\partial_\nu + m^2\big)\psi =& 0\\
\text{setting } \{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}.& \implies\\
(\partial^2 + m^2)\psi =& 0
\end{aligned}
$$

[1] defines the $\gamma$s in terms of the Pauli matrices $\sigma/\tau$:

$$
\begin{aligned}
\gamma^0 =& I \otimes \tau_3\\
\gamma^j =& i\sigma^j \otimes \tau_2
\end{aligned}
$$

### 2.1.1 Cousins of the Gamma Matrices

We collect some properties of the Pauli matrices.

$$\sigma^x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{1}$$

$$\sigma^y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \tag{2}$$

$$\sigma^z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{3}$$

$$
\begin{aligned}
\sigma^x\sigma^y\sigma^z =& \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\\
=& \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\\
=& iI
\end{aligned}
$$

The Dirac gamma matrices have the following representation.

$$\gamma^0 = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \tag{4}$$

$$\gamma^i = \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix} \tag{5}$$

From [1, II.1,(2)]

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \tag{6}$$

$$\{\gamma^0, \gamma^0\} = 2\eta^{00}$$

$$\gamma^0\gamma^0 = I \tag{7}$$

$$\{\gamma^i, \gamma^i\} = 2\eta^{ii}$$

$$\gamma^i\gamma^i = -I \tag{8}$$

Expanding (6)

$$\begin{aligned}
\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu &= 2\eta^{\mu\nu} \\
\gamma^\mu\gamma^\nu &= \eta^{\mu\nu} + \eta^{\mu\nu} - \gamma^\nu\gamma^\mu \\
&= \eta^{\mu\nu} - i\sigma^{\mu\nu}, \text{ where} \\
\sigma^{\mu\nu} &= i\big(\eta^{\mu\nu} - \gamma^\nu\gamma^\mu\big). \text{ Now apply (6) again} \\
&= i\big(\frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) - \gamma^\nu\gamma^\mu\big) \\
&= i\big(\frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)\big) \\
&= \frac{i}{2}[\gamma^\mu, \gamma^\nu]
\end{aligned}$$

From [1, II,(7)]

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 \tag{9}$$

Using the representation (4) and (5).

$$\begin{aligned}
\gamma^5 &= i \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \begin{bmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{bmatrix} \begin{bmatrix} 0 & \sigma^2 \\ -\sigma^2 & 0 \end{bmatrix} \begin{bmatrix} 0 & \sigma^3 \\ -\sigma^3 & 0 \end{bmatrix} \\
&= i \begin{bmatrix} 0 & \sigma^1 \\ \sigma^1 & 0 \end{bmatrix} \begin{bmatrix} -\sigma^2\sigma^3 & 0 \\ 0 & -\sigma^2\sigma^3 \end{bmatrix} \\
&= i \begin{bmatrix} 0 & -\sigma^1\sigma^2\sigma^3 \\ -\sigma^1\sigma^2\sigma^3 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}
\end{aligned}$$

Which agrees with [1, II,(8)].

$$\begin{aligned}
\gamma^0(\gamma^1\gamma^2\gamma^3) &= -\,i\gamma^5 \\
(\gamma^0)^2(\gamma^1\gamma^2\gamma^3) &= -\,i\gamma^0\gamma^5 \\
I\gamma^1\gamma^2\gamma^3 &= -\,i\gamma^0\gamma^5 \\
\gamma^1\gamma^2\gamma^3 &= -\,i\gamma^0\gamma^5
\end{aligned}$$

From (6):

$$\begin{aligned}
\gamma^0\gamma_i =& \eta^{0i} - \gamma^i\gamma^0 \\
=& -\,\gamma^i\gamma^0, \text{ so (9) becomes:} \\
\gamma^5 =& -\,i\gamma^1\gamma^0\gamma^2\gamma^3 \\
\gamma^1\gamma^5 =& -\,i(\gamma^1)^2\gamma^0\gamma^2\gamma^3 \\
i\gamma^0\gamma^2\gamma^3 =& \gamma^1\gamma^5, \text{ from (8)} \\
\gamma^0\gamma^2\gamma^3 =& -\,i\gamma^1\gamma^5
\end{aligned}$$

### 2.1.2 Isomorphism $\frac{SU(2)}{Z^2} \cong SO(3)$

In [2, Section 3] a $2 \times 2$ unimodular unitary matrix has the form:

$$\begin{aligned}
U =& \begin{bmatrix} x & -y \\ \bar{y} & \bar{x} \end{bmatrix}. \text{ Substituting} \\
x =& \cos\theta e^{i\phi},\ y = \sin\theta e^{i\psi}, \text{ gives} \\
U =& \begin{bmatrix} \cos\theta e^{i\phi} & \sin\theta e^{i\psi} \\ -sin\theta e^{-i\psi} & \cos\theta e^{-i\phi} \end{bmatrix} \\
U^T =& \begin{bmatrix} \cos\theta e^{-i\phi} & -\sin\theta e^{i\psi} \\ sin\theta e^{-i\psi} & \cos\theta e^{i\phi} \end{bmatrix}
\end{aligned} \tag{10}$$

We follow [2, Section 4] and transform the Pauli Matrices.

$$\begin{aligned}
U\sigma^1U^T &= \begin{bmatrix} \cos\theta e^{i\phi} & \sin\theta e^{i\psi} \\ -\sin\theta e^{-i\psi} & \cos\theta e^{-i\phi} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \cos\theta e^{-i\phi} & -\sin\theta e^{i\psi} \\ \sin\theta e^{-i\psi} & \cos\theta e^{i\phi} \end{bmatrix} \\
&= \begin{bmatrix} \sin\theta e^{i\psi} & \cos\theta e^{i\phi} \\ \cos\theta e^{-i\phi} & -\sin\theta e^{-i\psi} \end{bmatrix} \begin{bmatrix} \cos\theta e^{-i\phi} & -\sin\theta e^{i\psi} \\ \sin\theta e^{-i\psi} & \cos\theta e^{i\phi} \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta\sin\theta\left[e^{i(\psi-\phi)} + e^{i(\phi-\psi)}\right] & \cos^2\theta e^{2i\phi} - \sin^2\theta e^{2i\psi} \\ \cos^2\theta e^{-2i\phi} - \sin^2\theta e^{-2i\psi} & -\cos\theta\sin\theta\left[e^{i(\psi-\phi)} + e^{i(\phi-\psi)}\right] \end{bmatrix}
\end{aligned} \tag{11}$$

The real part of the off diagonal elements of (11) is:

$$\begin{aligned}
&\begin{bmatrix} 0 & \Re(\cos^2\theta e^{2i\phi} - \sin^2\theta e^{2i\psi}) \\ \Re(\cos^2\theta e^{-2i\phi} - \sin^2\theta e^{-2i\psi}) & 0 \end{bmatrix} \\
=&\frac{1}{2}(\cos^2\theta e^{2i\phi} - \sin^2\theta e^{2i\psi} + \cos^2\theta e^{-2i\phi} - \sin^2\theta e^{-2i\psi})\sigma^1 \\
=&\Re(x^2 - y^2)
\end{aligned}$$

The imaginary part of the off diagonal elements of (11) is:

$$\begin{aligned}
&\begin{bmatrix} 0 & \Im(\cos^2\theta e^{2i\phi} - \sin^2\theta e^{2i\psi}) \\ \Im(\cos^2\theta e^{-2i\phi} - \sin^2\theta e^{-2i\psi}) & 0 \end{bmatrix} \\
=&\frac{1}{2}(\cos^2\theta e^{2i\phi} - \sin^2\theta e^{2i\psi} - \cos^2\theta e^{-2i\phi} + \sin^2\theta e^{-2i\psi})\frac{\sigma^2}{i} \\
=&-\Im(x^2 - y^2)
\end{aligned}$$

The diagonal elements of (11) are:

$$\begin{aligned}
\cos\theta\sin\theta\left[e^{i(\psi-\phi)} + e^{i(\phi-\psi)}\right]\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} =& \cos\theta\sin\theta\left[e^{i(\psi-\phi)} + e^{i(\phi-\psi)}\right]\sigma^3, \text{ now} \\
\cos\theta\sin\theta e^{i(\psi-\phi)} =& \cos\theta e^{-i\phi}\sin\theta e^{i\psi} \\
=&\bar{x}y, \text{ and} \\
\cos\theta\sin\theta e^{i(\phi-\psi)} =& \cos\theta e^{i\phi}\sin\theta e^{-i\psi} \\
=&\bar{x}y, \text{ whence} \\
\cos\theta\sin\theta\left[e^{i(\psi-\phi)} + e^{i(\phi-\psi)}\right] =& x\bar{y} + \bar{x}y \\
=&2\Re x\bar{y}, \text{ and} \\
\cos\theta\sin\theta\left[e^{i(\psi-\phi)} + e^{i(\phi-\psi)}\right]\sigma^3 =& 2\Re(x\bar{y})\sigma^3
\end{aligned}$$

So (11) becomes:

$$U\sigma^1 U^T = \Re(x^2 - y^2)\sigma^1 - \Im(x^2 - y^2)\sigma^2 + 2\Re(x\bar{y})\sigma^3 \tag{12}$$

$$\begin{aligned}
U\sigma^2 U^T =& \begin{bmatrix} \cos\theta e^{i\phi} & \sin\theta e^{i\psi} \\ -\sin\theta e^{-i\psi} & \cos\theta e^{-i\phi} \end{bmatrix}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} \cos\theta e^{-i\phi} & -\sin\theta e^{i\psi} \\ \sin\theta e^{-i\psi} & \cos\theta e^{i\phi} \end{bmatrix} \\
=&i\begin{bmatrix} \cos\theta e^{i\phi} & \sin\theta e^{i\psi} \\ -\sin\theta e^{-i\psi} & \cos\theta e^{-i\phi} \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \cos\theta e^{-i\phi} & -\sin\theta e^{i\psi} \\ \sin\theta e^{-i\psi} & \cos\theta e^{i\phi} \end{bmatrix} \\
=&i\begin{bmatrix} \sin\theta e^{i\psi} & -\cos\theta e^{i\phi} \\ \cos\theta e^{-i\phi} & \sin\theta e^{-i\psi} \end{bmatrix}\begin{bmatrix} \cos\theta e^{-i\phi} & -\sin\theta e^{i\psi} \\ \sin\theta e^{-i\psi} & \cos\theta e^{i\phi} \end{bmatrix} \\
=&i\begin{bmatrix} \sin\theta e^{i\psi}\cos\theta e^{-i\phi} - \cos\theta e^{i\phi}\sin\theta e^{-i\psi} & -\sin^2\theta e^{2i\psi} - \cos^2\theta e^{2i\phi} \\ \cos^2\theta e^{-2i\phi} + \sin^2\theta e^{-2i\psi} & -\cos\theta e^{-i\phi}\sin\theta e^{i\psi} + \sin\theta e^{-i\psi}\cos\theta e^{i\phi} \end{bmatrix} \\
=&i\begin{bmatrix} \sin\theta\cos\theta\left[e^{i(\psi-\phi)} - e^{i(\phi-\psi)}\right] & -\sin^2\theta e^{2i\psi} - \cos^2\theta e^{2i\phi} \\ \cos^2\theta e^{-2i\phi} + \sin^2\theta e^{-2i\psi} & \sin\theta\cos\theta\left[-e^{i(\psi-\phi)} + e^{i(\phi-\psi)}\right] \end{bmatrix}
\end{aligned} \tag{13}$$

The real part of the off diagonal terms in (13) is:

$$
\begin{aligned}
&\begin{bmatrix} 0 & \Im(\sin^2\theta e^{2i\psi} + \cos^2\theta e^{2i\phi}) \\ -\Im(\cos^2\theta e^{-2i\phi} + \sin^2\theta e^{-2i\psi}) & 0 \end{bmatrix} \\
=&\begin{bmatrix} 0 & (\sin^2\theta \sin 2i\psi + \cos^2\theta \sin 2i\phi) \\ -(\cos^2\theta \sin(-2i\phi) + \sin^2\theta \sin(-2i\psi)) & 0 \end{bmatrix} \\
=&\begin{bmatrix} 0 & (\sin^2\theta \sin 2i\psi + \cos^2\theta \sin 2i\phi) \\ (\cos^2\theta \sin(2i\phi) + \sin^2\theta \sin(2i\psi)) & 0 \end{bmatrix} \\
=&(\sin^2\theta \sin 2i\psi + \cos^2\theta \sin 2i\phi)\sigma^1 \\
=&\Im(x^2+y^2)\sigma^1
\end{aligned}
$$

The imaginary part of the off diagonal terms in (13) is:

$$
\begin{aligned}
\begin{bmatrix} 0 & -\Re(\sin^2\theta e^{2i\psi} + \cos^2\theta e^{2i\phi}) \\ \Re(\cos^2\theta e^{-2i\phi} + \sin^2\theta e^{-2i\psi}) & 0 \end{bmatrix} =& \begin{bmatrix} 0 & -\Re(x^2+y^2) \\ \Re(\bar{x}^2+\bar{y}^2) & 0 \end{bmatrix} \\
=& \frac{\Re(x^2+y^2)}{i}\sigma_2
\end{aligned}
$$

The diagonal terms in (13) are:

$$
\begin{aligned}
i\sin\theta\cos\theta\big[e^{i(\psi-\phi)} - e^{i(\phi-\psi)}\big]\sigma^3 =& i\big[\bar{x}y - x\bar{y}\big]\sigma^3 \\
=& i2i\Im(\bar{x}y)\sigma^3 \\
=& -2\Im(\bar{x}y)\sigma^3
\end{aligned}
$$

So (13) becomes:

$$
U\sigma^2U^T = \Im(x^2+y^2)\sigma^1 + \Re(x^2+y^2)\sigma_2 - 2\Im(\bar{x}y)\sigma^3 \tag{14}
$$

$$
\begin{aligned}
U\sigma^3U^T &= \begin{bmatrix} \cos\theta e^{i\phi} & \sin\theta e^{i\psi} \\ -\sin\theta e^{-i\psi} & \cos\theta e^{-i\phi} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos\theta e^{-i\phi} & -\sin\theta e^{i\psi} \\ \sin\theta e^{-i\psi} & \cos\theta e^{i\phi} \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta e^{i\phi} & -\sin\theta e^{i\psi} \\ -\sin\theta e^{-i\psi} & -\cos\theta e^{-i\phi} \end{bmatrix} \begin{bmatrix} \cos\theta e^{-i\phi} & -\sin\theta e^{i\psi} \\ \sin\theta e^{-i\psi} & \cos\theta e^{i\phi} \end{bmatrix} \\
&= \begin{bmatrix} \cos^2\theta - \sin^2\theta & -2\cos\theta\sin\theta e^{i(\theta+\phi)} \\ -2\cos\theta\sin\theta e^{-i(\theta+\phi)} & -(\cos^2\theta - \sin^2\theta) \end{bmatrix}
\end{aligned} \tag{15}
$$

Now

$$
\begin{aligned}
xy &= \cos\theta e^{i\phi}\sin\theta e^{i\psi} \\
&= \cos\theta\sin\theta e^{i(\phi+\psi)} \\
&= \cos\theta\sin\theta\big[\cos(\phi+\psi) + i\sin(\phi+\psi)\big] \\
&= \cos\theta\sin\theta\big[\cos[-(\phi+\psi)] - i\sin[-(\phi+\psi)]\big]
\end{aligned}
$$

We see that the real part of the cross term in (15) is:

$$\begin{aligned}
-2\cos\theta\sin\theta\begin{bmatrix}0 & \Re(e^{i(\phi+\psi)})\\ \Re(e^{-i(\phi+\psi)}) & 0\end{bmatrix} &= -2\cos\theta\sin\theta\begin{bmatrix}0 & \Re(e^{i(\phi+\psi)})\\ \Re(e^{-i(\phi+\psi)}) & 0\end{bmatrix}\\
&= -2\cos\theta\sin\theta\begin{bmatrix}. & \Re(e^{i(\phi+\psi)})\\ \Re(e^{i(\phi+\psi)}) & .\end{bmatrix}\\
&= -2\cos\theta\sin\theta\Re(e^{i(\phi+\psi)})\sigma^1\\
&= 2\Re(xy)\sigma^1
\end{aligned}$$

Then the imaginary part of the cross term in (15) is:

$$\begin{aligned}
-2\cos\theta\sin\theta\begin{bmatrix}0 & \Im(e^{i(\phi+\psi)})\\ \Im(e^{-i(\phi+\psi)}) & 0\end{bmatrix} &= -2\cos\theta\sin\theta\begin{bmatrix}0 & \Im(e^{i(\phi+\psi)})\\ -\Im(e^{i(\phi+\psi)}) & 0\end{bmatrix}\\
&= 2\cos\theta\sin\theta\Im(e^{i(\phi+\psi)})\frac{\sigma^2}{i}\\
&= 2\Im(xy)\frac{\sigma^2}{i}
\end{aligned}$$

Finally

$$\begin{aligned}
&|x|^2-|y|^2 = \cos^2\theta-\sin^2\theta, \text{ so the diagonal term in (15) is}\\
&(|x|^2-|y|^2)\sigma^3
\end{aligned}$$

So (15) becomes:

$$U\sigma^3U^T = 2\Re(xy)\sigma^1 + 2\Im(xy)\sigma^2 + (|x|^2-|y|^2)\sigma^3 \tag{16}$$

Collecting (12), (14), and (16), we get:

$$\begin{aligned}
U\sigma^1U^T &= \Re(x^2-y^2)\sigma^1 - \Im(x^2-y^2)\sigma^2 + 2\Re(x\bar{y})\sigma^3\\
U\sigma^2U^T &= \Im(x^2+y^2)\sigma^1 + \Re(x^2+y^2)\sigma_2 - 2\Im(\bar{x}y)\sigma^3\\
U\sigma^3U^T &= 2\Re(xy)\sigma^1 + 2\Im(xy)\sigma^2 + (|x|^2-|y|^2)\sigma^3
\end{aligned} \tag{17}$$

## 2.2 Quantizing the Dirac Field

## 2.3 Lorentz Group and Weyl Spinors

The generators of the Lorentz Group are defined in [1, II.3,(2,5,6)]:

$$J_1 = \begin{bmatrix}0&0&0&0\\0&0&0&0\\0&0&0&-i\\0&0&i&0\end{bmatrix}\quad J_2 = \begin{bmatrix}0&0&0&0\\0&0&0&i\\0&0&0&0\\0&-i&0&0\end{bmatrix}\quad J_3 = \begin{bmatrix}0&0&0&0\\0&0&-i&0\\0&i&0&0\\0&0&0&0\end{bmatrix} \tag{18}$$

$$iK_1 = \begin{bmatrix}0&1&0&0\\1&0&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix}\quad iK_2 = \begin{bmatrix}0&0&1&0\\0&0&0&0\\1&0&0&0\\0&0&0&0\end{bmatrix}\quad iK_3 = \begin{bmatrix}0&0&0&1\\0&0&0&0\\0&0&0&0\\1&0&0&0\end{bmatrix} \tag{19}$$

**Theorem 1.**

$$[J_i, J_j] = i\epsilon_{ijk} J_k \tag{20}$$
$$[J_i, K_j] = i\epsilon_{ijk} K_k \tag{21}$$
$$[K_i, K_j] = -\, i\epsilon_{ijk} J_k \tag{22}$$

*Proof.* From (18):

$$\begin{aligned}
[J_1, J_2] &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
&= iJ_3
\end{aligned}$$

From (19):

$$\begin{aligned}
[K_1, K_2] &= -\,[iK_1, iK_2] \\
&= [iK_2, iK_1] \\
&= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
&= -\, iJ_3, \text{ using (18).}
\end{aligned}$$

□

**Theorem 2** ($J_+$ and $J_-$ form two separate $SU(2)$ algebras)**.** *If we define:*

$$J_{\pm i} \equiv \frac{1}{2}\big(J_i \pm iK_i\big), \textit{ then}$$
$$\big[J_{\pm i}, J_{\pm j}\big] = i\epsilon_{ijk} J_{\pm k}, \textit{ and} \tag{23}$$
$$\big[J_{+i}, J_{-j}\big] = 0 \tag{24}$$

*Proof.*

$$
\begin{aligned}
\big[J_{+i}, J_{+j}\big] =& \frac{1}{4}\big[J_i + iK_i, J_j + iK_j\big] \\
=& \frac{1}{4}\big[\big(J_i + iK_i\big)\big(J_j + iK_j\big) - \big(J_j + iK_j\big)\big(J_i + iK_i\big)\big] \\
=& \frac{1}{4}\big[J_iJ_j + iJ_iK_j + iK_iJ_j - K_iK_j - J_jJ_i - iJ_jK_i - iK_jJ_i + K_jK_i\big] \\
=& \frac{1}{4}\big[[J_i, J_j] - [K_i, K_j] + i[J_i, K_j] - i[J_j, K_i]\big] \\
=& \frac{1}{4}\big[i\epsilon_{ijk}J_k + i\epsilon_{ijk}J_k + i^2\epsilon_{ijk}K_k - i^2\epsilon_{jik}K_k]\big], \text{ from (20),(21),(22)} \\
=& \frac{i}{2}\big[\epsilon_{ijk}J_k + i\epsilon_{ijk}K_k\big] \\
=& \frac{i}{2}\epsilon_{ijk}J_{+k}
\end{aligned}
$$

Which is the + case of (23). Define

$$
\begin{aligned}
K'_i =& -K_i \\
J'_{+i} =& J_i + K'_i
\end{aligned}
$$

We see that the argument for (23) suffices to establish:

$$
\begin{aligned}
\big[J'_{+i}, J'_{+j}\big] =& \frac{i}{2}\epsilon_{ijk}J'_{+k}, \text{ but} \\
J'_{+i} =& J_{-i}, \text{ whence} \\
\big[J_{-i}, J_{-j}\big] =& \frac{i}{2}\epsilon_{ijk}J_{-k}
\end{aligned}
$$

Which is the − case of (23). Finally:

$$
\begin{aligned}
\big[J_{+i}, J_{-j}\big] =& \frac{1}{4}\big[J_i + iK_i, J_j - iK_j\big] \\
=& \frac{1}{4}\big[\big(J_i + iK_i\big)\big(J_j - iK_j\big) - \big(J_j - iK_j\big)\big(J_i + iK_i\big)\big] \\
=& \frac{1}{4}\big[J_iJ_j - iJ_iK_j + iK_iJ_j + K_iK_j - J_jJ_i - iJ_jK_i + iK_jJ_i - K_jK_i\big] \\
=& \frac{1}{4}\big[[J_i, J_j] + [K_i, K_j] - i[J_i, K_j] + i\underbrace{[K_i, J_j]}_{-[J_j, K_i]}\big] \\
=& \frac{1}{4}\big[\cancel{i\epsilon_{ijk}J_k} - \cancel{i\epsilon_{ijk}J_k} - \cancel{i^2\epsilon_{ijk}K_k} - \cancel{i^2\epsilon_{jik}K_k}\big], \text{ from (20),(21),(22)} \\
=& 0
\end{aligned}
$$

Which establishes (24). □

## 2.4 Spin-Statistics Connection

## 2.5 Vacuum Energy, Grassman Integrals, and Feynman Diagrams for Fermions

## 2.6 Electron Scattering and Gauge Invariance

## 2.7 Diagrammatic Proof of Gauge Invariance

## 2.8 Photon-Electron Scattering

## References


[1] Anthony Zee. *Quantum Field Theory in a Nutshell*. Princeton University Press, 2010.

[2] Dennis B Westra. SU(2) and SO(3). `https://www.mat.univie.ac.at/~westra/so3su2.pdf`, 2008. [Online; accessed 4-December-2021].